

Quiz

Mar. 30, 2026

Which of the followings is/are true about page tables?

- A** A single page table is shared across all processes in the system to reduce memory overhead.
- B** In a two-level page table, the page table itself occupies a single contiguous block of physical memory regardless of how many mappings exist.
- C** The valid bit in a PTE indicates whether a virtual page has a valid mapping to a physical page frame.
- D** Multi-level page tables are primarily used to reduce the amount of physical memory consumed by page tables, especially for sparse virtual address spaces.
- E** The dirty bit in a PTE is used to indicate whether the page has been modified since it was loaded into memory.

提交

Which of the following statements about the Translation Lookaside Buffer (TLB) is/are true?

- A** The TLB is a software-managed cache that stores recently used page table entries
- B** On a TLB hit, the page frame number is obtained directly from the TLB without accessing the page table in memory.
- C** Flushing the TLB on every context switch is one way to prevent address mapping conflicts between different processes.
- D** The TLB's effectiveness relies heavily on spatial and temporal locality of memory accesses.
- E** A TLB miss always results in a page fault.

提交

Which of the following statements about page table size is/are true?

- A** For a 32-bit virtual address space with 4KB pages and 4-byte PTEs, a single-level page table would require 4MB of memory per process.
- B** Multi-level page tables increase the total page table size compared to single-level page tables because they add indexing overhead at each level.
- C** Using larger page sizes (e.g., 16KB instead of 4KB) decreases the number of PTEs needed to cover a given virtual address space.
- D** The size of a page table entry varies depending on which level of the page table it belongs to in a multi-level structure.
- E** In a multi-level page table, only the page table pages that correspond to valid virtual memory regions need to be allocated in physical memory.

提交

Which of the following statements about demand paging are true?

- A** Demand paging loads pages from disk to memory only when they are accessed, rather than pre-loading all pages at process startup.
- B** When a page fault occurs and no free page frames are available, the OS must perform page replacement to evict an existing page.
- C** In demand paging, all pages (including code pages from program binaries) must be written back to disk when evicted from memory.
- D** The present bit in a page table entry indicates whether the page is currently in physical memory or stored on disk.
- E** On a page fault, the hardware automatically handles the entire process of fetching the page from disk and updating the page table.

提交

Which of the following statements about demand paging are true?

- A** When a page fault occurs, the hardware automatically fetches the missing page from disk and updates the page table entry before resuming execution.
- B** The page fault handler first identifies the faulting virtual address by reading it from a register.
- C** If no free page frames are available during page fault handling, the OS must select a victim page for replacement based on a page replacement policy.
- D** When a dirty page is selected for replacement, the OS must write its contents back to disk before reusing the page frame.
- E** After the page fault is handled and the page table is updated, the CPU resumes execution by executing the next instruction.

提交

多选题 2分

If memory access latency is 500ns, and the page fault handling time is 5ms, if the effective access time leads to a slowdown of less than 100%, the page fault rate is approximately _____ .

- ☐ A 1%
- ☐ B 0.1%
- ☐ C 0.01%
- ☐ D 0.001%

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